

HAWKEYE SYSTEM: 3D RECONSTRUCTION AND SFM

Presented by Fady Damak – YV9GAA

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The Problem

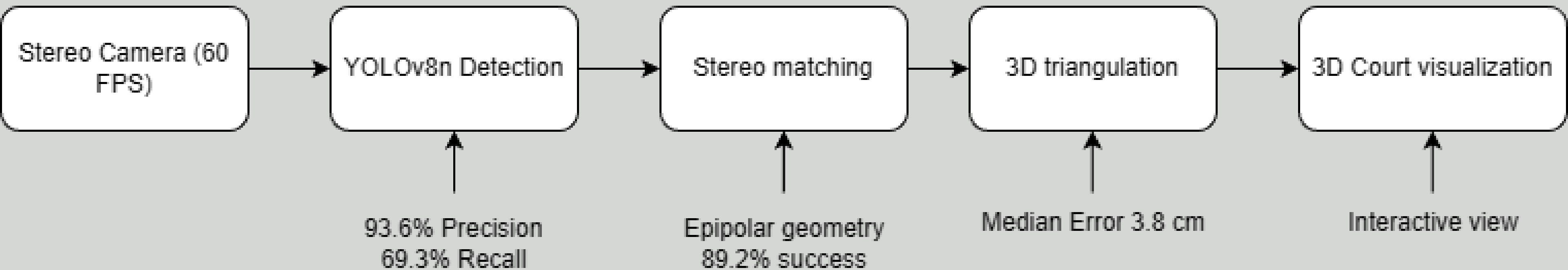
- Hawk-Eye costs \$60–70k USD
- Methods and algorithms are not publicly available
- Technology inaccessible to amateur clubs

The Solution

- Stereo Vision + YOLOv8 Detection
- Mid-range computer (GTX 1650)
- Open Source Software

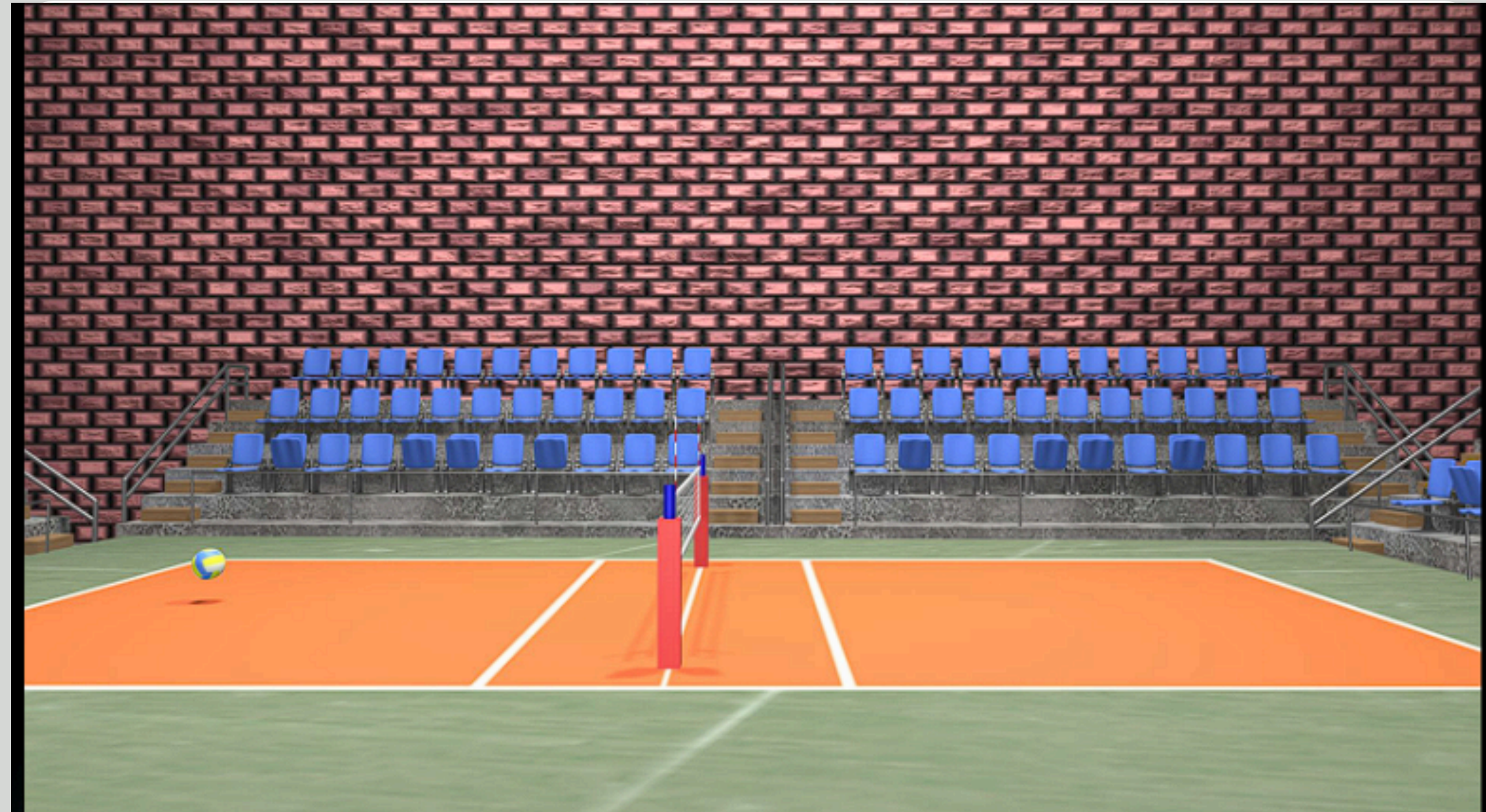


System Architecture



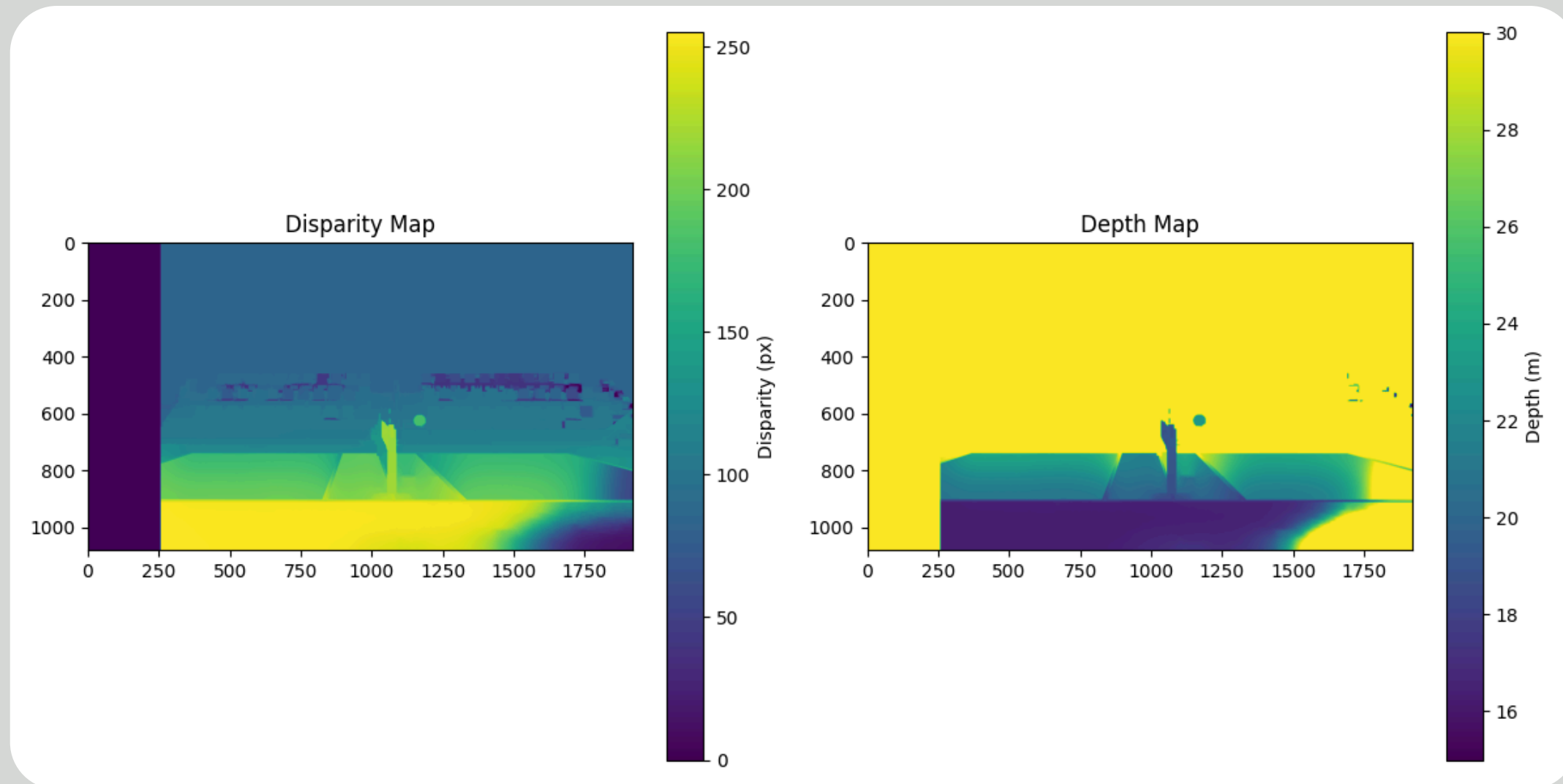
Technical Implementation

- **Ball Detection:** YOLOv8n trained on 39,750 images (Roboflow + Blender)
- **Stereo Matching:** Semi-Global Block Matching (SGBM) with epipolar geometry
- **3D Reconstruction:** Triangulation using calibrated camera matrices
- **Visualization:** Homography mapping to top-down court view + 3D point cloud



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Key Challenges

Challenge 01 Data acquisition

- No recording equipment available
- The footage has to be synchronized between the 2 cameras
- **Built a synthetic dataset in Blender**
- **Manual synchronization using sliders**

Challenge 02 Domain Gap

- Domain Gap in AI model training
- Performance dropped to 31.5% on synthetic Blender data
- **Mixed-domain training: 31.5% → 79.9% mAP50**

Challenge 03 Camera calibration

- Camera calibration is required (for extrinsic and intrinsic parameters)
- **Checkerboard board + manual parameter tuning**
- **30 minutes setup time, technical expertise needed**
- **Barrier to adoption by non-expert users**

Camera & Court Configuration

Camera ParametersCourt ParametersStereo Matching

Camera Hardware

Focal Length (mm):

30.0

Sensor Width (mm):

36.0

Resolution Width (px):

1920

Resolution Height (px):

1080

Camera Setup

Baseline (m):

3.0

Distance between cameras

Depth Range

Min Distance (m):

0.0

Closest trackable distance

Max Distance (m):

50.0

Farthest trackable distance

Calculated Values

Focal Length (px):

1600.00

Reset to Defaults

Cancel

Save & Apply

hawkeye

FPS N/A | GPU 16% | CPU 26% | LAT N/A

Chapter 5: Results and Validation.md | thesis.md | Chapter 6: Discussion.md | ball_positions_camera.csv | measure_hardv

Volleyball Tracking System

Camera & Court Configuration

Video Frame Extractor

Frame Analyzer

Process Complete Videos

v1.0.0

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666605027,0.894230766364645,15.38461533530572

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BUG CONSOLE | TERMINAL | PORTS | SQL HISTORY | TASK MONITOR | GITLENS

thesis

tools

env

gitignore

blender_model_comparison.csv

blender_model_comparison.json

OUTLINE

TIMELINE

0: 384x640 1 Volleyball, 38.9ms

Speed: 2.2ms preprocess, 38.9ms inference, 0.7ms postprocess per image at shape (1, 3, 384, 640)

Ball coordinates in left image: 1302, 657

Ball coordinates in 3D (det-based): 3.8863636216425625, 1.3295454495092978, 18.18181811294766

Camera to World: [3.8864 1.3295 18.182] → [-0.11825 -2.6131 1.7964]

Visualizing 99 ball positions

Results exported to F:/hawkeye/test-vids

powershell

Python

Col 4: 2 | Ln 4, Col 58 | Spaces: 4 | UTF-8 | CRLF | {} CSV

main*

Launchpad

0

1

Live Share

CSVLint

Query

Align

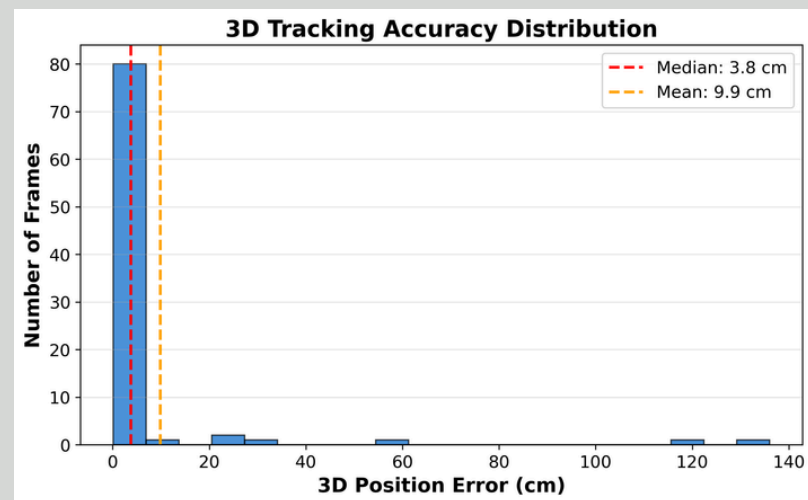
Windows Taskbar

System Tray

Results

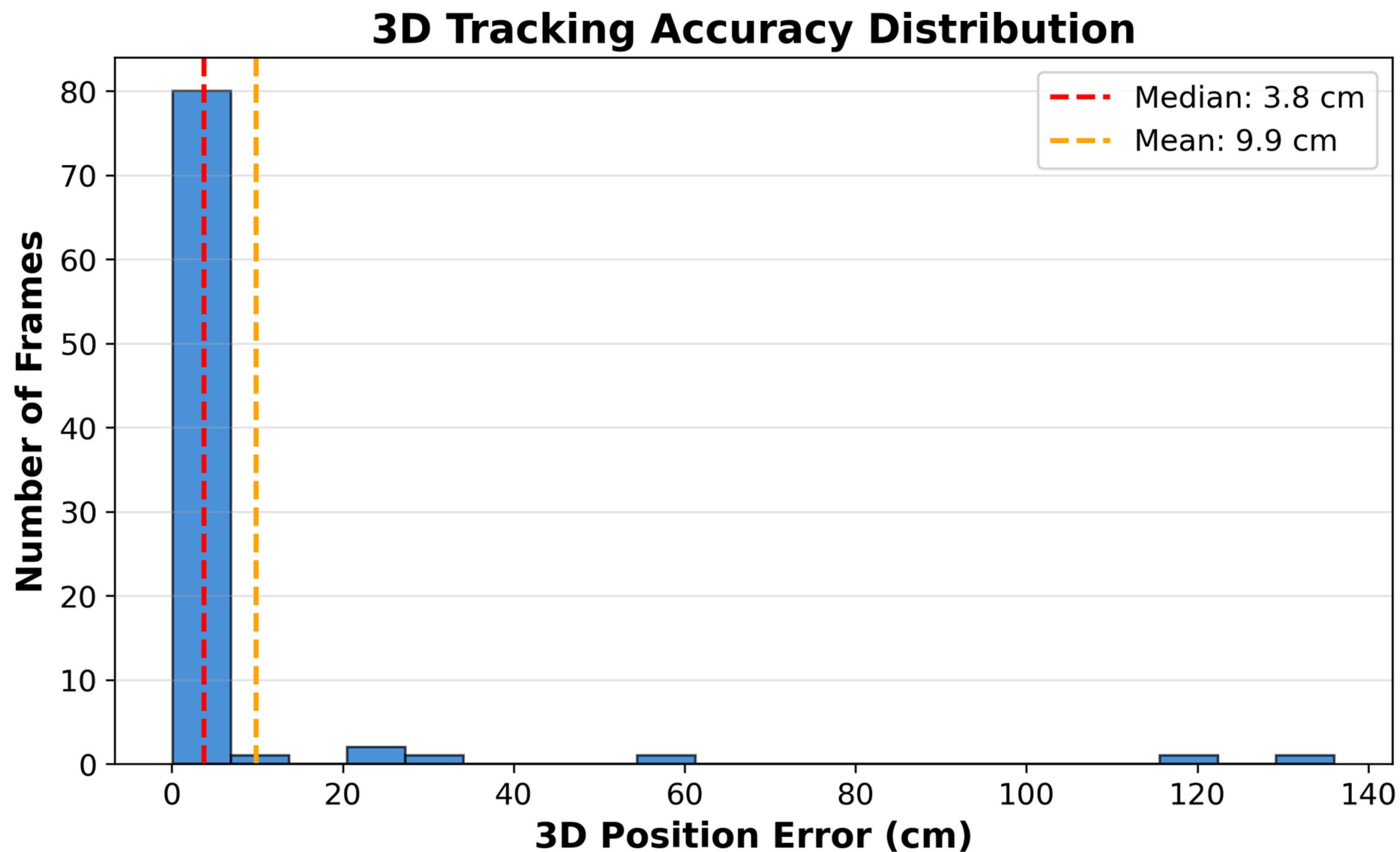
Accuracy

- Median: 3.8 cm
- Mean: 9.9cm
- 95th percentile: 41 cm



Performance

- Processing: 9.7 FPS
- Success Rate: 87%
- Detection Precision: 89.6%
- Detection Recall: 73%



Limitations

Validation

Validated only on Blender
synthetic footage
Real-world testing is a
logical next step
Domain gap uncertainty
remains

Hardware

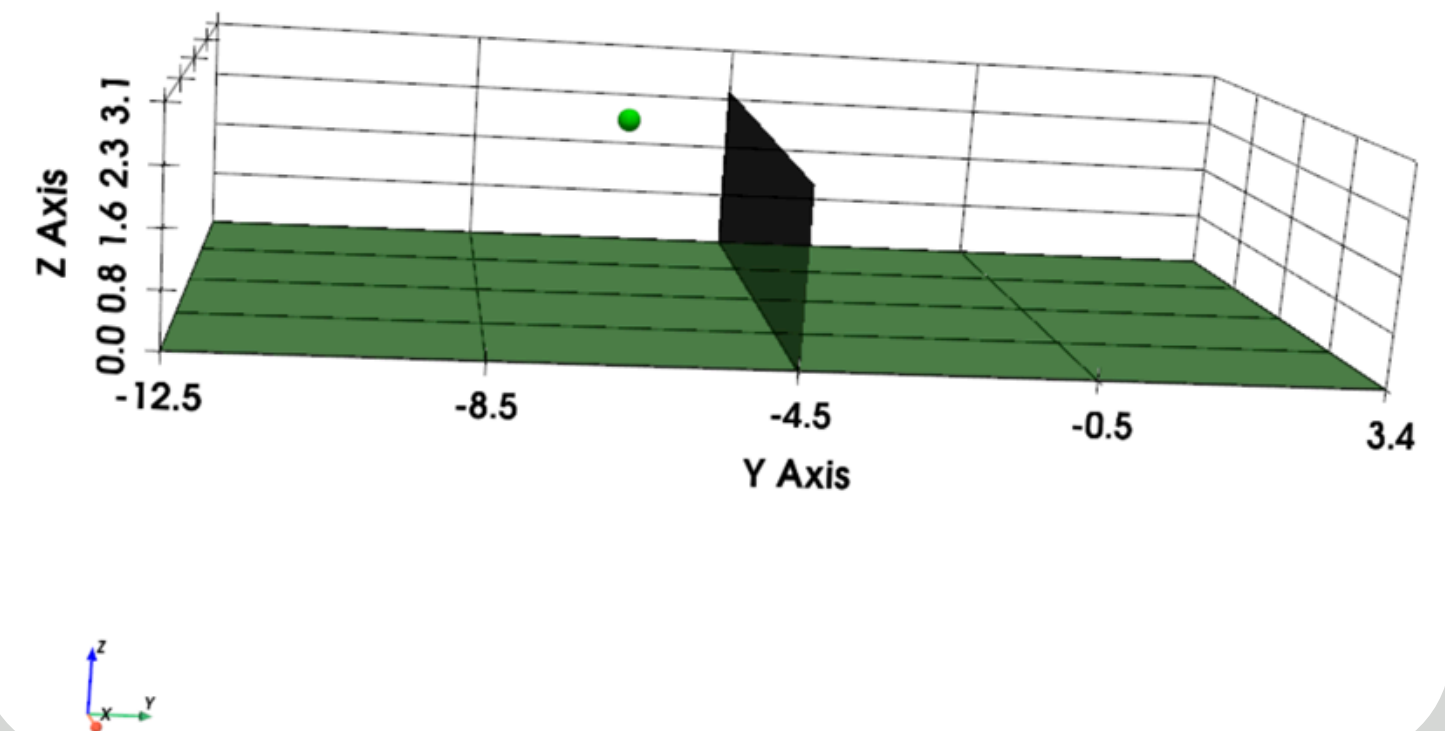
This system requires a
discrete GPU like GTX 1650
or equivalent, which isn't
found on budget laptops

Analysis

For now, the application is
for post-match analysis
only, with 9.7 FPS
performance only (should
be 30 for live capability)

Summary

- Centimeter-level 3D tracking (median error 3.8 cm accuracy)
- Consumer hardware accessible (~\$300–700 for cameras + GPU vs \$60–70K Hawk-Eye system)
- 87% pipeline success rate (validated on 200 frames of footage)



THANK you

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